

Faiyaj Rahman

U.S. Citizen | 586-455-7796 | faiyajr@umich.edu | [linkedin.com/in/faiyajrahman](https://www.linkedin.com/in/faiyajrahman) | github.com/faiyajr | faiyajr.dev

EDUCATION

University of Michigan

Ann Arbor, MI

B.S.E. in Computer Engineering | GPA: 3.6/4.0 · Dean's List

Expected Graduation: May 2028

- **Coursework:** Data Structures & Algorithms, Computer Architecture, Object Oriented Programming, Discrete Math, Computing Systems, Linear Algebra, Multivariable & Vector Calculus

HONORS

Jane Street FOCUS | Selected from 1000+ applicants for quant trading and SWE program

May 2026

WORK EXPERIENCE

Bosch

Farmington Hills, MI

Software Engineering Intern

May 2026 – Aug. 2026

- Deployed a Flask operations platform to AWS and scripted multi-tenant KPIs and analytics scaling to **5000+** devices.
- Designed a Python tool to convert CAN .asc logs to MF4 with DBC channel mapping, saving **20+** engineer-hours monthly.
- Shipped a 3-agent pipeline, cleaning and tagging internal documents for a master agent, cutting new-hire escalations **40%**.
- Engineered a multi-threaded GUI tool used by **100+** developers to build Python executables, saving **\$15K** in licenses.

Hikmah Hackers

Warren, MI

Co-Founder & Lead Instructor

Sep. 2026 – Present

- Co-founded a coding and mentorship program for underserved middle and high school students at an Islamic STEM school, teaching programming fundamentals through problem sets & hands-on mini-projects, and giving students career guidance.
- Designing a curriculum spanning CS topics, game/web dev, and AI scripting, featuring hackathons offering cash prizes.

Saf · Darul Uloom Michigan

Warren, MI

Software Developer

Nov. 2025 – Present

- Launched a school-management beta on Postgres, an Express REST API, and React featuring role-based access control.
- Implemented assignments, grading, and attendance, replacing paper grades for **20+** educators in a **3-month** live pilot.
- Driving AI integration on a 3-engineer team, scoping LLM report card narratives & role-scoped natural language querying.

MRacing FSAE

Ann Arbor, MI

Software Engineer

Aug. 2025 – Present

- Built a C++ Raspberry Pi dashboard decoding **1000+** CAN messages/sec into driver analytics & fault states over **3** pages.
- Wrote firmware for an infrared lap trigger, edge-detecting photosensor voltage drops on beam breaks to timestamp laps.

University of Michigan Electric Vehicle Center

Ann Arbor, MI

Undergraduate Research Software Developer

Sep. 2025 – May 2026

- Developed open-source Python software for an EV charging module, contributing to **\$10k** in net savings for an OEM.
- Designed a Simulink FSM for system-safety logic and charging protocols, porting it to Python for pipeline integration.

PROJECTS

Overround | C++, Python, PyTorch, React, D3.js, FastAPI, PostgreSQL

Jul. 2026 – Present

- Building a transformer with learned club embeddings over **18,000+** matches to predict UEFA Champions League outcomes, benchmarking against Dixon-Coles and Elo.
- Engineering a multithreaded C++ Monte Carlo engine simulating the **36-team** league phase at **5,000+** sims per sec.

Limit Order Book & Matching Engine | C++, Linux, Git, gTest

May 2026 – Present

- Building a C++ limit order book with price-time priority at **10K–100K+** orders/sec and **sub-millisecond** latency.
- Optimizing with cache-efficient data structures and custom allocators, cutting latency **~60%** over **1M+** order events.

Low-Level Modulator | Verilog, Assembly, FPGA, Digital Logic

Jan. 2026 – Apr. 2026

- Engineered an **18-instruction** processor in Verilog on a DE2-115 FPGA driving a synthesizer with **40** assembly modules.
- Designed an integer-math DSP pipeline synthesizing **32-bit** audio from **3** wavetables with **5** modulations at **30 FPS**.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL, Lua, Julia, Bash, Assembly, Verilog, MATLAB

Tools/Frameworks: Claude Code/Cursor Skills, React, Node/Express, PyTorch, Tailwind, PySide6, Flask, GoogleTest, PostgreSQL, Supabase, AWS, Docker, Kubernetes, Linux, GitLab, CAN, Raspberry Pi, Simulink

Practices: Agile/Scrum, Git/GitHub, CI/CD, code review, unit testing, RAG & LLM agents